



Basic Electronics-SEE-Sep-oct 2023 copy

Basic Electronics (Visvesvaraya Technological University)



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CBCS SCHEME

BBEE103/203

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2023 Basic Electronics for EEE Stream

Max. Marks: 100

Time: 3 hrs.

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.
3. Assume missed data.

Module - 1			M	L	C
Q.1	a.	Explain forward and reverse characteristics of semiconductor diode.	8	L2	CO1
	b.	Calculate forward and reverse resistances offered by a silicon diode with $I_f = 100\text{mA}$ at $V_R = 50\text{V}$. Assume V_F for silicon diode to be 0.75V and reverse current $I_R \approx 100\text{nA}$.	4	L3	CO1
	c.	What is piecewise linear characteristic? With neat diagram explain diode approximation of ideal diode and practical diode.	8	L2	CO1
OR					
Q.2	a.	Describe the working of full wave bridge rectifier.	8	L2	CO1
	b.	Explain zener diode as voltage regulator with no load and with load.	6	L2	CO1
	c.	Illustrate RC- π filter.	6	L2	CO1
Module - 2					
Q.3	a.	Explain output characteristics of a transistor in common base configuration.	6	L2	CO2
	b.	Describe the procedure for drawing dc - load line on transistor CE output characteristics.	8	L2	CO2
	c.	Calculate I_c and I_B for a transistor that has $\alpha_{DC} = 0.98$, $I_B = 100\mu\text{A}$. Also determine the value of β_{DC} for the transistor.	6	L3	CO2
OR					
Q.4	a.	Explain common Emitter input characteristics. $V_{BE} \approx 0.7\text{V}$	6	L2	CO2
	b.	Explain how transistor can be used as current amplifier.	6	L2	CO2
	c.	Explain the working of N-channel JFET.	8	L2	CO2

Module – 3						
Q.5	a.	Explain Inverting and Non-inverting amplifier.	8	L2	CO2	
	b.	Define Op-Amp. Mention any 5 ideal characteristics of an op-amp.	6	L2	CO2	
	c.	Draw a summer circuit with $V_1 = +1V$, $V_2 = +3V$, $V_3 = +2V$, $R_1 = R_2 = R_3 = 2K\Omega$. Determine the output voltage when $R_F = 3K\Omega$.	6	L3	CO2	
OR						
Q.6	a.	Explain the working of op-amp as Differentiator.	8	L2	CO2	
	b.	Define : i) Input offset current ii) Input bias current iii) slew rate iv) CMRR	6	L2	CO2	
	c.	With block diagram, explain basic structure of an Op amp. Also write its equivalent circuit diagram.	6	L2	CO2	
Module – 4						
Q.7	a.	Convert the following : i) $(2AB.8)_{16} = ()_{10}$ ii) $(416.12)_{10} = ()_8$ iii) $(25.375)_{10} = ()_2$ iv) $(16.2)_8 = ()_{16}$	6	L2	CO3	
	b.	Find complement of the function i) $F_1 = x'yz' + x'y'z$ ii) $F_2 = x(y'z' + yz)$ Using De-Morgan's theorem.	8	L2	CO3	
	c.	Explain the working of Half adder.	6	L2	CO3	
OR						
Q.8	a.	Express the Boolean function $F = A + B/C$ in sum of minterms.	6	L3	CO3	
	b.	Mention the postulates and theorems of Boolean algebra.	8	L2	CO3	
	c.	Explain the working of full adder.	6	L2	CO3	
Module – 5						
Q.9	a.	Describe the working of LVDT.	6	L2	CO4	
	b.	Explain the working principle of capacitive pressure transducer.	6	L2	CO4	
	c.	With neat block diagram, explain the working of communication system.	8	L2	CO4	
OR						
Q.10	a.	Describe a Thermistor and sketch approximate resistance/temperature characteristics for a thermistor.	6	L2	CO1	
	b.	Write a short notes of photo diodes	6	L2	CO5	
	c.	What is modulation? Describe the need of modulation in communication system.	8	L2	CO5	